

[745] (CONTINUED)

ON THE SCHWARZIAN DERIVATIVE AND THE POLYHEDRAL FUNCTIONS.

**111.** Selecting the transformations which correspond to the positive substitutions  $abcd$ , and completing the group of 24 we have

*Homographic Transformations. The groups of 12 and 24. Pole at A.*

|    | $(A\xi + B) \div (C\xi + D)$   |                                 |                                         |                                        |         |
|----|--------------------------------|---------------------------------|-----------------------------------------|----------------------------------------|---------|
| 1  | 1                              | 0                               | 0                                       | 1                                      | 1       |
| 2  | 0                              | -1                              | 1                                       | 0                                      | $ab.cd$ |
| 3  | $\epsilon + \epsilon^4$        | $\epsilon^4$                    | $\epsilon^4$                            | $-(\epsilon + \epsilon^4)$             | $ac.bd$ |
| 4  | $-\epsilon^3$                  | $\epsilon^2 + \epsilon^4$       | $\epsilon^2 + \epsilon^4$               | $\epsilon^3$                           | $ab.cd$ |
| 5  | $-\epsilon$                    | $\epsilon + \epsilon^2$         | $\epsilon + \epsilon^2$                 | $\epsilon$                             | $acb$   |
| 6  | $-\epsilon$                    | $\epsilon$                      | $\epsilon^2$                            | $-(\epsilon + \epsilon^2)$             | $abc$   |
| 7  | $\epsilon + \epsilon^4$        | $\epsilon^4$                    | $\epsilon$                              | $-(\epsilon + \epsilon^4)$             | $acd$   |
| 8  | $\epsilon^2 + 1$               | 1                               | 1                                       | $-(\epsilon + \epsilon^2)$             | $bcd$   |
| 9  | $\epsilon + \epsilon^2$        | $\epsilon$                      | $\epsilon$                              | $-(\epsilon + \epsilon^4)$             | $bdc$   |
| 10 | $\epsilon^2 + 1$               | $1 + \epsilon^4$                | $1 + \epsilon^3$                        | -1                                     | $abdc$  |
| 11 | -1                             | $1 + \epsilon^2$                | $1 + \epsilon^3$                        | -1                                     | $ab$    |
| 12 | $-\epsilon^2$                  | $1 + \epsilon^4$                | $1 + \epsilon^2$                        | 1                                      | $bd$    |
| 13 | 1                              | $1 + 2\epsilon^4$               | $1 + 2\epsilon$                         | -1                                     | $ac$    |
| 14 | $-\epsilon^2 + \epsilon^3$     | $1 + \epsilon + 3\epsilon^4$    | $-1 - 3\epsilon - \epsilon^4$           | $\epsilon^2 - \epsilon^3$              | $cd$    |
| 15 | $\epsilon^2 - \epsilon^3$      | $3 + \epsilon + \epsilon^2$     | $-1 - 3\epsilon - \epsilon^2$           | $-\epsilon^2 + \epsilon^3$             | $ac$    |
| 16 | $-1 + \epsilon^2$              | $-1 - \epsilon^2 + 2\epsilon^4$ | $1 + \epsilon^2 - 2\epsilon^4$          | $1 - \epsilon^2$                       | $bd$    |
| 17 | $2 + \epsilon^2 + 2\epsilon^4$ | $-2 - 2\epsilon^3 - \epsilon^4$ | $2\epsilon + \epsilon^2 + 2\epsilon^3$  | $2\epsilon + 2\epsilon^3 + \epsilon^4$ | $ad$    |
| 18 | $2 + 2\epsilon^3 + \epsilon^4$ | $2 + \epsilon^2 + 2\epsilon^3$  | $-2\epsilon - 2\epsilon^2 - \epsilon^3$ | $2\epsilon + \epsilon^2 + 2\epsilon^4$ | $bc$    |
| 19 | $-2 + \epsilon + \epsilon^4$   | $-\epsilon + \epsilon^4$        | $-\epsilon + \epsilon^4$                | $\epsilon + \epsilon^4 - 2\epsilon^2$  | $abcd$  |
| 20 | 1                              | -1                              | 1                                       | 1                                      | $abdc$  |
| 21 | 1                              | 1                               | -1                                      | 1                                      | $acdb$  |
| 22 | $1 + \epsilon + 3\epsilon^2$   | $\epsilon^3 - \epsilon^2$       | $\epsilon^3 - \epsilon^2$               | $1 + 3\epsilon + \epsilon^2$           | $acbd$  |
| 23 | $1 + 2\epsilon^2$              | -1                              | -1                                      | $-1 - 2\epsilon$                       | $adbc$  |
| 24 | $3 + \epsilon + \epsilon^2$    | $-\epsilon^3 + \epsilon^4$      | $-\epsilon^2 + \epsilon^4$              | $\epsilon + 3\epsilon + \epsilon^4$    | $adcb$  |

As an example of the calculation we have  $(A, B, C, D) = (0, i, -1, 0)$ ;  $ab$ . Hence

$$\begin{aligned} A_1, B_1, C_1, D_1 &= (a(i-1), b(i-1) + i + 1, b(i-1) - (i+1), -a(i+1)), \\ &= \left(1, \frac{b-i}{a}, \frac{b+i}{a}, -1\right). \end{aligned}$$

The second and third coefficients are

$$\frac{\sqrt{5}+1}{2} - i\sqrt{\frac{5+\sqrt{5}}{2}}, \quad \frac{\sqrt{5}+1}{2} + i\sqrt{\frac{5+\sqrt{5}}{2}},$$

which, in virtue of the values of  $\epsilon$  and  $\epsilon^4$ , are  $1 + 2\epsilon^4$  and  $1 + 2\epsilon$  respectively: or the result is as above  $(1, 1 + 2\epsilon^4, 1 + 2\epsilon, -1)$ .

**112.** In like manner for the passage from the  $\Theta$ -form to the  $B$ -form, if  $X, Y, Z$  be the coordinates of a point on the spherical surface in regard to the  $\Theta$ -axes,  $X_1, Y_1, Z_1$  those of the same point in regard to the  $B$ -axes, we may write

$$X : Y : Z = X_1 : bY_1 + aZ_1 : -aY_1 + bZ_1,$$

where

$$a, b = \frac{\sqrt{5}-1}{2\sqrt{3}}, \frac{\sqrt{5}+1}{2\sqrt{3}}.$$

Hence  $X : Y : Z = L : M : N$ , being the equations of an axis of rotation in the first set of coordinates, those of the same axis in the second set of coordinates will be

$$X_1 : bY_1 + aZ_1 : -aY_1 + bZ_1 = L : M : N,$$

or calling these

$$= X_1 : Y_1 : Z_1 = L_1 : M_1 : N_1,$$

we have

$$L_1, M_1, N_1 = L, bM - aN, aM + bN;$$

these values are such that

$$L_1^2 + M_1^2 + N_1^2 = L^2 + M^2 + N^2,$$

or  $\lambda, \mu, \nu, \lambda_1, \mu_1, \nu_1$  being the rotations, we have

$$L, M, N = \vartheta \lambda, \vartheta \mu, \vartheta \nu; \quad L_1, M_1, N_1 = \vartheta \lambda_1, \vartheta \mu_1, \vartheta \nu_1,$$

where  $\vartheta$  has the same value in the two sets of equations. We have thus

$$B + C : B - C : D - A : D + A = L : 2M : N : -i\vartheta,$$

$$B_1 + C_1 : B_1 - C_1 : D_1 - A_1 : D_1 + A_1 = L_1 : 2M_1 : N_1 : -i\vartheta;$$

and hence

$$\begin{aligned} B_1 + C_1 &= B + C, \\ B_1 - C_1 &= b(B - C) - ai(D - A), \\ D_1 - A_1 &= -ai(B - C) + b(D - A), \\ D_1 + A_1 &= D + A; \end{aligned}$$

and thence

$$\begin{aligned} A_1 &= ai(B - C) - b(D - A) + (D + A), \\ B_1 &= b(B - C) - ai(D - A) + (B + C), \\ C_1 &= -b(B - C) + ai(D - A) + (B + C), \\ D_1 &= -ai(B - C) + b(D - A) + (D + A). \end{aligned}$$

**113.** As an example of the transformation, take

$$(A, B, C, D) = (2, -3 + \sqrt{5} + i(1 - \sqrt{5}), -3 + \sqrt{5} + i(-1 + \sqrt{5}), -2) \quad [bc.de];$$

then

$$B - C, B + C, D - A, D + A = i(1 - \sqrt{5}), -3 + \sqrt{5}, -2, 0;$$

and thence

$$\begin{aligned} A_1 &= \frac{1}{2\sqrt{3}} i(-6 + 2\sqrt{5}) + \frac{1}{2\sqrt{3}} (-2 + 2\sqrt{5}), \\ B_1 &= \frac{1}{2\sqrt{3}} (-4i) + \frac{1}{2\sqrt{3}} (2i(1 + \sqrt{5}) + (-3 + \sqrt{5})), \\ C_1 &= \frac{1}{2\sqrt{3}} (4i) + \frac{1}{2\sqrt{3}} (2i(1 - \sqrt{5}) + (-3 + \sqrt{5})), \\ D_1 &= \frac{1}{2\sqrt{3}} (-6 + 2\sqrt{5}) + \frac{1}{2\sqrt{3}} (-2 - 2\sqrt{5}); \end{aligned}$$

viz. multiplying by  $2\sqrt{3}$ , these are

$$8, i(-6 + 2\sqrt{5}) + 2\sqrt{3}(-3 + \sqrt{5}), i(6 - 2\sqrt{5}) + 2\sqrt{3}(-3 + \sqrt{5}), -8,$$

that is,

$$8, (-6 + 2\sqrt{5})(i + \sqrt{3}), (-6 + 2\sqrt{5})(-i + \sqrt{3}), -8,$$

or since

$$2 + \sqrt{3} = -2i\omega \quad \text{and} \quad -2 + \sqrt{3} = 2i\omega^2,$$

dividing by 4 these are

$$2, i(3 - \sqrt{5})\omega, i(-3 + \sqrt{5})\omega^2, -2,$$

as in the table.

**114.** *Homographic Transformations. The group of 60. Pole at B.*

$$\omega = \frac{1}{2}(-1 + i\sqrt{3}).$$

|    |                         | $(A\xi + B) \div (C\xi + D)$      |                                   |                                   |         |
|----|-------------------------|-----------------------------------|-----------------------------------|-----------------------------------|---------|
|    |                         | 0                                 | 0                                 | 1                                 | 1       |
| 1  | 1                       |                                   |                                   |                                   |         |
| 2  | 1                       | $i(3 - \sqrt{5})$                 | $i(-3 + \sqrt{5})$                | -2                                | $ac.bd$ |
| 3  | 0                       | $\omega$                          | 1                                 | 0                                 | $ae.bd$ |
| 4  | 0                       | $\omega^2$                        | 1                                 | 0                                 | $bd.ce$ |
| 5  | 2                       | $i(3 - \sqrt{5})$                 | $i(-3 + \sqrt{5})$                | -2                                | $ab.cd$ |
| 6  | 2                       | $i(-3 - \sqrt{5})$                | $i(3 + \sqrt{5})$                 | -2                                | $ad.bc$ |
| 7  | 2                       | $i(3 - \sqrt{5})\omega$           | $i(-3 + \sqrt{5})\omega^2$        | -2                                | $be.cd$ |
| 8  | 2                       | $i(-3 - \sqrt{5})\omega$          | $i(3 + \sqrt{5})\omega^2$         | -2                                | $ad.be$ |
| 9  | 2                       | $i(-3 - \sqrt{5})\omega^2$        | $i(3 + \sqrt{5})\omega$           | -2                                | $ab.ce$ |
| 10 | 2                       | $i(3 - \sqrt{5})\omega^2$         | $i(-3 + \sqrt{5})\omega$          | -2                                | $ae.bc$ |
| 11 | 2                       | $(-\sqrt{3} - i\sqrt{5})$         | $(-\sqrt{3} + i\sqrt{5})\omega^2$ | -2                                | $ac.de$ |
| 12 | 2                       | $(-\sqrt{3} - i\sqrt{5})\omega^2$ | $-\sqrt{3} + i\sqrt{5}$           | -2                                | $ad.ce$ |
| 13 | 2                       | $(\sqrt{3} - i\sqrt{5})\omega$    | $(-\sqrt{3} + i\sqrt{5})\omega$   | -2                                | $ae.cd$ |
| 14 | 2                       | $(\sqrt{3} - i\sqrt{5})\omega^2$  | $\sqrt{3} + i\sqrt{5}$            | -2                                | $ac.de$ |
| 15 | 2                       | $(\sqrt{3} - i\sqrt{5})$          | $(\sqrt{3} + i\sqrt{5})\omega^2$  | -2                                | $ad.ce$ |
| 16 | 2                       | $(\sqrt{3} - i\sqrt{5})\omega$    | $(\sqrt{3} + i\sqrt{5})\omega^2$  | -2                                | $ae.cd$ |
| 17 | $\omega$                | 0                                 | 0                                 | 1                                 | $ace$   |
| 18 | $\omega^2$              | 0                                 | 0                                 | 1                                 | $aec$   |
| 19 | $\sqrt{3} - i\sqrt{5}$  | 2                                 | -2                                | $\sqrt{3} + i\sqrt{5}$            | $bcd$   |
| 20 | $-\sqrt{3} - i\sqrt{5}$ | $2\omega^2$                       | $-2\omega$                        | $-\sqrt{3} + i\sqrt{5}$           | $bde$   |
| 21 | $-\sqrt{3} - i\sqrt{5}$ | $2\omega$                         | $-2\omega^2$                      | $-\sqrt{3} + i\sqrt{5}$           | $bdc$   |
| 22 | $\sqrt{3} - i\sqrt{5}$  | $2\omega$                         | $-2\omega^2$                      | $\sqrt{3} + i\sqrt{5}$            | $bcd$   |
| 23 | $\sqrt{3} - i\sqrt{5}$  | $2\omega^2$                       | $-2\omega$                        | $\sqrt{3} + i\sqrt{5}$            | $bdc$   |
| 24 | $\sqrt{3} - i\sqrt{5}$  | $2\omega^2$                       | $-2\omega$                        | $\sqrt{3} + i\sqrt{5}$            | $abd$   |
| 25 | $2\omega$               | $-\sqrt{3} - i\sqrt{5}$           | $-\sqrt{3} + i\sqrt{5}$           | $-2\omega^2$                      | $adb$   |
| 26 | $-\sqrt{3} - i\sqrt{5}$ | $-\sqrt{3} - i\sqrt{5}$           | $-\sqrt{3} + i\sqrt{5}$           | $\sqrt{3} + i\sqrt{5}$            | $abc$   |
| 27 | $-\sqrt{3} - i\sqrt{5}$ | $-\sqrt{3} - i\sqrt{5}\omega^2$   | $(-\sqrt{3} + i\sqrt{5})\omega$   | $\sqrt{3} + i\sqrt{5}$            | $acb$   |
| 28 | $\sqrt{3} - i\sqrt{5}$  | $-\sqrt{3} - i\sqrt{5}\omega$     | $(-\sqrt{3} + i\sqrt{5})\omega^2$ | $\sqrt{3} + i\sqrt{5}$            | $abe$   |
| 29 | $\sqrt{3} - i\sqrt{5}$  | $-\sqrt{3} - i\sqrt{5}\omega^2$   | $(-\sqrt{3} + i\sqrt{5})\omega$   | $\sqrt{3} + i\sqrt{5}$            | $aeb$   |
| 30 | $2\omega^2$             | $-\sqrt{3} - i\sqrt{5}$           | $(-\sqrt{3} + i\sqrt{5})\omega^2$ | $-2\omega$                        | $acd$   |
| 31 | $\omega$                | $\sqrt{3} - i\sqrt{5}$            | $\sqrt{3} + i\sqrt{5}$            | $-2\omega^2$                      | $adc$   |
| 32 | $2\omega$               | $\sqrt{3} - i\sqrt{5}$            | $\sqrt{3} + i\sqrt{5}$            | $-2\omega^2$                      | $ade$   |
| 33 | $\omega^2$              | $\sqrt{3} - i\sqrt{5}$            | $\sqrt{3} + i\sqrt{5}$            | -2                                | $aed$   |
| 34 | $2\omega^2$             | $\sqrt{3} - i\sqrt{5}$            | $(\sqrt{3} + i\sqrt{5})\omega^2$  | $-2\omega$                        | $bce$   |
| 35 | $\omega^2$              | $\sqrt{3} - i\sqrt{5}$            | $(\sqrt{3} + i\sqrt{5})\omega^2$  | $-2\omega^2$                      | $bec$   |
| 36 | 2                       | $\sqrt{3} - i\sqrt{5}$            | $(-\sqrt{3} + i\sqrt{5})\omega$   | $-2\omega^2$                      | $cde$   |
| 37 | $2\omega$               | $-\sqrt{3} - i\sqrt{5}$           | $(-\sqrt{3} + i\sqrt{5})\omega$   | $-2\omega$                        | $ced$   |
| 38 | $2\omega^2$             | $-\sqrt{3} - i\sqrt{5}$           | $(-\sqrt{3} + i\sqrt{5})\omega$   | -2                                | $adceb$ |
| 39 | $2\omega^2$             | 0                                 | $(\sqrt{3} + i\sqrt{5})\omega$    | 0                                 | $adceb$ |
| 40 | 0                       | $\sqrt{3} - i\sqrt{5}$            | $(\sqrt{3} + i\sqrt{5})\omega$    | $-2\omega^2$                      | $acbde$ |
| 41 | 2                       | 0                                 | 0                                 | $-2\omega$                        | $aedbc$ |
| 42 | $2\omega$               | $\sqrt{3} - i\sqrt{5}$            | $i(-3 + \sqrt{5})$                | 0                                 | $abecd$ |
| 43 | $2\omega$               | 0                                 | -2                                | -2                                | $aedcb$ |
| 44 | 0                       | $i(3 - \sqrt{5})\omega^2$         | 0                                 | -2                                | $adbec$ |
| 45 | $2\omega$               | 0                                 | $-2\omega$                        | 0                                 | $acebd$ |
| 46 | 0                       | $+2\omega^2$                      | 0                                 | $-2\omega^2$                      | $abcde$ |
| 47 | 2                       | 0                                 | $i(-3 + \sqrt{5})\omega$          | 0                                 | $adebc$ |
| 48 | 0                       | 2                                 | $i(-3 + \sqrt{5})$                | $-2\omega^2$                      | $aecdb$ |
| 49 | 2                       | $i(3 - \sqrt{5})$                 | -2                                | 0                                 | $abdce$ |
| 50 | 0                       | 0                                 | 0                                 | $(-\sqrt{3} - i\sqrt{5})\omega^2$ | $acbed$ |

|    | $(A\xi + B) \div (C\xi + D)$ |                           |                            |                                  |              |
|----|------------------------------|---------------------------|----------------------------|----------------------------------|--------------|
| 51 | $-\sqrt{3} - i\sqrt{5}$      | $i(3 - i\sqrt{5})\omega$  | $-2\omega^2$               | $(\sqrt{3} + i\sqrt{5})\omega$   | <i>acdeb</i> |
| 52 | 0                            | 0                         | 0                          | 0                                | <i>adbce</i> |
| 53 | $\sqrt{3} - i\sqrt{5}$       | $2\omega$                 | $i(-3 + \sqrt{5})\omega^2$ | $-2\omega$                       | <i>aecbd</i> |
| 54 | 0                            | 0                         | $i(-3 + \sqrt{5})\omega^2$ | $-2\omega$                       | <i>abedc</i> |
| 55 | 2                            | 2                         | 0                          | 0                                | <i>aebcd</i> |
| 56 | 2                            | $i(3 - \sqrt{5})$         | $-2\omega^2$               | $(-\sqrt{3} + i\sqrt{5})\omega$  | <i>abdec</i> |
| 57 | $-\sqrt{3} - i\sqrt{5}$      | 0                         | $-2\omega$                 | $(\sqrt{3} + i\sqrt{5})\omega^2$ | <i>acedb</i> |
| 58 | 0                            | $i(3 - \sqrt{5})\omega^2$ | 0                          | 0                                | <i>aebdc</i> |
| 59 | $\sqrt{3} - i\sqrt{5}$       | 0                         | $i(-3 + \sqrt{5})\omega$   | $-2\omega^2$                     | <i>adcbe</i> |
| 60 | 0                            | $2\omega^2$               | $i(3 + \sqrt{5})\omega$    | $-2\omega$                       | <i>adecb</i> |

115. We hence derive

*Homographic Transformations. The groups of 12 and 24. Pole at B.*

|    | $(A\xi + B) \div (C\xi + D)$                 |                                              |                                              |                                              |              |
|----|----------------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|--------------|
| 1  | 1                                            | 0                                            | 0                                            | 1                                            | 1            |
| 2  | 2                                            | $i(3 - \sqrt{5})$                            | $i(-3 + \sqrt{5})$                           | -2                                           | <i>ab.cd</i> |
| 3  | 0                                            | 1                                            | 1                                            | 0                                            | <i>ac.bd</i> |
| 4  | 2                                            | $i(-3 - \sqrt{5})$                           | $i(3 + \sqrt{5})$                            | -2                                           | <i>ad.bc</i> |
| 5  | $2\omega$                                    | $-\sqrt{3} - i\sqrt{5}$                      | $\sqrt{3} + i\sqrt{5}$                       | $-2\omega^2$                                 | <i>abc</i>   |
| 6  | $2\omega^2$                                  | $-\sqrt{3} - i\sqrt{5}$                      | $\sqrt{3} + i\sqrt{5}$                       | $-2\omega$                                   | <i>acb</i>   |
| 7  | $-\sqrt{3} - i\sqrt{5}$                      | $2\omega$                                    | $-2\omega^2$                                 | $-\sqrt{3} + i\sqrt{5}$                      | <i>abe</i>   |
| 8  | $\sqrt{3} - i\sqrt{5}$                       | $\sqrt{3} - i\sqrt{5}$                       | $\sqrt{3} + i\sqrt{5}$                       | $\sqrt{3} + i\sqrt{5}$                       | <i>aeb</i>   |
| 9  | $\omega^2$                                   | $\sqrt{3} - i\sqrt{5}$                       | $-2\omega$                                   | $-\sqrt{3} + i\sqrt{5}$                      | <i>abd</i>   |
| 10 | $\omega$                                     | $2\omega^2$                                  | $-2\omega$                                   | $-\sqrt{3} + i\sqrt{5}$                      | <i>adb</i>   |
| 11 | $\sqrt{3} - i\sqrt{5}$                       | $2\omega$                                    |                                              |                                              | <i>acd</i>   |
| 12 | $-\sqrt{3} - i\sqrt{5}$                      |                                              |                                              |                                              | <i>adc</i>   |
| 13 | 0                                            | $\sqrt{3}(1 + \sqrt{5}) + i(-3 - \sqrt{5})$  | $\sqrt{3}(1 + \sqrt{5}) + i(3 + \sqrt{5})$   | -2                                           | <i>ab</i>    |
| 14 | 0                                            | $\sqrt{3}(-1 - \sqrt{5}) + i(-3 - \sqrt{5})$ | $\sqrt{3}(-1 - \sqrt{5}) + i(3 + \sqrt{5})$  | -2                                           | <i>cd</i>    |
| 15 | $\sqrt{5}$                                   | -i                                           | i                                            | $-\sqrt{5}$                                  | <i>ae</i>    |
| 16 | $i\sqrt{5}$                                  | $i\sqrt{5}$                                  | $-i\sqrt{5}$                                 | -1                                           | <i>bd</i>    |
| 17 | 0                                            | $\sqrt{3}(-1 + \sqrt{5}) + i(3 - \sqrt{5})$  | $\sqrt{3}(-1 + \sqrt{5}) + i(-3 + \sqrt{5})$ | -2                                           | <i>ad</i>    |
| 18 | 0                                            | $\sqrt{3}(1 - \sqrt{5}) + i(3 - \sqrt{5})$   | $\sqrt{3}(1 - \sqrt{5}) + i(-3 + \sqrt{5})$  | -2                                           | <i>bc</i>    |
| 19 | 0                                            | i                                            | i                                            | 1                                            | <i>abcd</i>  |
| 20 | 0                                            | -i                                           | -i                                           | 1                                            | <i>adcb</i>  |
| 21 | $\sqrt{3}(1 - \sqrt{5}) + i(3 + \sqrt{5})$   | 2                                            | -2                                           | $\sqrt{3}(1 - \sqrt{5}) + i(-3 + \sqrt{5})$  | <i>abde</i>  |
| 22 | $\sqrt{3}(1 + \sqrt{5}) + i(-3 + \sqrt{5})$  | 2                                            | -2                                           | $\sqrt{3}(1 + \sqrt{5}) + i(3 + \sqrt{5})$   | <i>acbd</i>  |
| 23 | $\sqrt{3}(-1 + \sqrt{5}) + i(3 - \sqrt{5})$  | 2                                            | -2                                           | $\sqrt{3}(-1 + \sqrt{5}) + i(-3 + \sqrt{5})$ | <i>acdb</i>  |
| 24 | $\sqrt{3}(-1 - \sqrt{5}) + i(-3 - \sqrt{5})$ | 2                                            | -2                                           | $\sqrt{3}(-1 - \sqrt{5}) + i(3 + \sqrt{5})$  | <i>adbc</i>  |

116. I give also the group of 12, (*abce*), slightly modifying the form: viz. I write first  $\sqrt{3} \pm \sqrt{5} = 2\sqrt{2}k$ , and therefore  $\sqrt{3} \mp \sqrt{5} = 2\sqrt{2}\frac{i}{k}$ ; then for  $x$  I write  $\lambda x$ , and divide the  $A$  and  $B$  by  $\lambda$ : the  $A$  and  $B$  then contain  $\frac{\lambda}{k}$ , and the  $C$  and  $D$  contain  $\frac{k}{\lambda}$ ; assuming  $\frac{\lambda}{k} = \frac{k}{\lambda}$ , we have  $\lambda = k$ . For instance, in the transformation corresponding to *abc*, the  $Ax + B$  and  $Cx + D$ ,

$$= 2\omega^2 x - (\sqrt{3} + i\sqrt{5}) \quad \text{and} \quad (-\sqrt{3} + i\sqrt{5})x - 2\omega,$$

become first  $2\omega^2 x - 2\sqrt{2}k$ , and  $-2\sqrt{2}ix - 2\omega$ , and then (omitting also the factor 2)

$$\omega^2 x - \sqrt{2}k \quad \text{and} \quad -\sqrt{2}\frac{i}{k}x - \omega,$$

viz. when  $k = i$ , they are  $\omega^2 x - i\sqrt{2}$  and  $-\omega x \cdot i\sqrt{2}$ ; that is, the values of  $A, B, C, D$  are  $\omega^2, -i\sqrt{2}, i\sqrt{2}, -\omega$ . The group is

|                                |            |                      |                     |             |                |
|--------------------------------|------------|----------------------|---------------------|-------------|----------------|
| <i>Group of 12. Pole at B.</i> | 1          | 0                    | 0                   | 1           | 1              |
|                                | $\omega$   | 0                    | 0                   | 1           | <i>ace</i>     |
|                                | $\omega^2$ | 0                    | 0                   | 1           | <i>aec</i>     |
|                                | $\omega$   | $-i\omega\sqrt{2}$   | $i\omega\sqrt{2}$   | $-\omega^2$ | <i>abc</i>     |
|                                | $\omega^2$ | $-i\omega^2\sqrt{2}$ | $i\omega^2\sqrt{2}$ | $-\omega$   | <i>acb</i>     |
|                                | 1          | $-i\omega\sqrt{2}$   | $i\sqrt{2}$         | $-\omega$   | <i>abe</i>     |
|                                | 1          | $-i\sqrt{2}$         | $i\omega\sqrt{2}$   | $-\omega^2$ | <i>aeb</i>     |
|                                | 1          | $-i\omega^2\sqrt{2}$ | $i\sqrt{2}$         | $-\omega^2$ | <i>bce</i>     |
|                                | 1          | $-i\sqrt{2}$         | $i\omega^2\sqrt{2}$ | $-\omega$   | <i>bec</i>     |
|                                | 0          | $-i\omega\sqrt{2}$   | $i\omega^2\sqrt{2}$ | 0           | <i>ab . ce</i> |
|                                |            |                      |                     |             | <i>ac . be</i> |
|                                | 0          | $-i\omega^2\sqrt{2}$ | $i\omega\sqrt{2}$   | 0           | <i>ae . bc</i> |
|                                | 0          | $-i\sqrt{2}$         | $i\sqrt{2}$         | 0           |                |

117. From the Table of the Groups of 12 and 24,  $\Theta$ -form, it appears that the group of 12 is

$$1, \frac{1}{x}, -1, -\frac{1}{x}, \frac{i(x-1)}{x+1}, \frac{-i(x-1)}{x+1}, \frac{i(x+1)}{x-1}, \frac{-i(x+1)}{x-1},$$

$$\frac{x+i}{x-i}, \frac{x-i}{x+i}, \frac{-(x+i)}{-x-i}, \frac{-(x-i)}{x+i},$$

and if we proceed to form the product of the twelve factors  $s - x$ ,  $s + \frac{1}{x}$ ,  $s + x$ , &c., we have first the three products

$$s^4 + \alpha s^2 + 1; \quad s^4 + \beta s^2 + 1; \quad s^4 + \gamma s^2 + 1;$$

if for shortness

$$\alpha, \beta, \gamma = -\left(x^2 + \frac{1}{x^2}\right), \frac{x^4 + 6x^2 + 1}{(x^2 - 1)^2}, -\frac{x^4 - 6x^2 + 1}{(x^2 + 1)^2}.$$

The product of the three quartic functions is

$$= (s^4 + 1)^3 + (s^4 + 1)^2 s^2 (\alpha + \beta + \gamma) + (s^4 + 1) s^4 (\beta\gamma + \gamma\alpha + \alpha\beta) + s^6 \cdot \alpha\beta\gamma,$$

and we have

$$\alpha + \beta + \gamma = -\frac{32x^2(x^4 + 1)}{(x^4 - 1)^2}, \quad \beta\gamma + \gamma\alpha + \alpha\beta = -\frac{36x^4}{(x^4 - 1)^2} = -\frac{4(x^8 - 33x^6 - 33x^2 + 1)}{x^2(x^4 - 1)^2},$$

$$\alpha\beta\gamma = \frac{4(x^8 - 33x^6 - 33x^2 + 1)}{x^2(x^4 - 1)^2}.$$

Hence the product is found to be

$$= (x^8 - 33x^6 - 33x^2 + 1) \div x^2(x^4 - 1)^2, \quad x^8 - 33x^6 - 33x^2 + 1$$

which is

$$= s^{12}(x^4 - 1)^2 \left\{ \frac{s^8 - 33s^6 - 33s^2 + 1}{x^2(x^4 - 1)^2} - \frac{x^8 - 33x^6 - 33x^2 + 1}{x^2(x^4 - 1)^2} \right\}.$$

We thus verify that the twelve transformations  $x$  into  $x$ , into  $\frac{1}{x}$ , &c., give each of them a transformation of the function

$$\frac{x^8 - 33x^6 - 33x^2 + 1}{x^2(x^4 - 1)^2}$$

into itself.

**118.** It has been already remarked that we can from the coefficients  $(A, B, C, D)$  of the homographic transformation pass back to the position of the axis of rotation: viz. we have

$$A, B, C, D = -i, -i\mu, -i\nu, +i;$$

and thence

$$\lambda : \mu : \nu : 1 = B + C : -i(B - C) : B - A : i(D + A),$$

that is,

$$\lambda, \mu, \nu = -i(B + C), -(B - C), -i(D - A); \frac{1}{2}(D + A).$$

The equations of the axis thus are

$$\frac{x}{B + C} : \frac{iy}{B - C} : \frac{z}{D - A},$$

and the equations of the central plane at right angles to the axis are

$$-(B + C)x + i(B - C)y + (A - D)z = 0.$$

**119.** In particular, we may find the equations of the 15 planes at right angles to the  $\Theta$ -axes: these are in fact the before-mentioned 15 planes, intersecting the sphere in great circles the projections of which are the circles in the three figures respectively. Taking the equation of the plane to be  $Lx + My + Nz = 0$ , it is at once seen that the equation of the projecting cone (vertex at the South pole) is

$$N(x^2 + y^2 + z^2 - 1) - 2(z + 1)(Lx + My + Nz) = 0,$$

and hence, writing  $z = 0$ , we find

$$N(x^2 + y^2 - 1) - 2(Lx + My) = 0$$

for the equation of the circle in the plane figure. We have thus the equations of a system of 15 circles related to each other in the manner before referred to.

**120.** Taking the  $\Theta$ -form, the equations of the 15 planes are at once found: and we thence obtain the equations of the 15 circles: viz. writing for shortness

$$\Omega = x^2 + y^2 - 1,$$

the equations are

|                                                 |           |                                                       |
|-------------------------------------------------|-----------|-------------------------------------------------------|
| $z = 0,$                                        | $(ab.cd)$ | $\Omega = 0,$                                         |
| $x = 0,$                                        | $(ac.bd)$ | $x = 0,$                                              |
| $y = 0,$                                        | $(ad.bc)$ | $y = 0,$                                              |
| $(3 - \sqrt{5})x + (1 - \sqrt{5})y + 2z = 0,$   | $(ae.bc)$ | $\Omega - [(3 - \sqrt{5})x + (1 - \sqrt{5})y] = 0,$   |
| $(-1 - \sqrt{5})x + (-1 + \sqrt{5})y + 2z = 0,$ | $(ab.ce)$ | $\Omega - [(-1 - \sqrt{5})x + (-1 + \sqrt{5})y] = 0,$ |
| $(1 + \sqrt{5})x + (3 + \sqrt{5})y + 2z = 0,$   | $(ac.be)$ | $\Omega - [(1 + \sqrt{5})x + (3 + \sqrt{5})y] = 0,$   |
| $(-3 + \sqrt{5})x + (-1 + \sqrt{5})y + 2z = 0,$ | $(ad.bc)$ |                                                       |
| $(1 + \sqrt{5})x + (1 - \sqrt{5})y + 2z = 0,$   | $(ab.de)$ |                                                       |
| $(-1 - \sqrt{5})x + (-3 - \sqrt{5})y + 2z = 0,$ | $(ae.bd)$ |                                                       |
| $(-3 + \sqrt{5})x + (1 - \sqrt{5})y + 2z = 0,$  | $(ad.ce)$ |                                                       |
| $(1 + \sqrt{5})x + (-1 + \sqrt{5})y + 2z = 0,$  | $(ae.cd)$ |                                                       |
| $(-1 - \sqrt{5})x + (3 + \sqrt{5})y + 2z = 0,$  | $(ac.de)$ |                                                       |
| $(3 - \sqrt{5})x + (-1 + \sqrt{5})y + 2z = 0,$  | $(bc.de)$ |                                                       |
| $(-1 - \sqrt{5})x + (1 - \sqrt{5})y + 2z = 0,$  | $(bc.cd)$ |                                                       |
| $(1 + \sqrt{5})x + (-3 - \sqrt{5})y + 2z = 0,$  | $(bd.ce)$ |                                                       |

and similarly for the other circles.

**121.** Observe that the arrangement is in sets of 3 planes, or circles, intersecting at right angles. One of the circles is the circle  $\Omega$ ,  $x^2 + y^2 - 1 = 0$ , corresponding to the equator, and two of them are the right lines  $x = 0$  and  $y = 0$ . The equations of the remaining 12 circles may be written in the somewhat different form

$$\begin{aligned}\Omega + (\sqrt{5} - 1) [y - \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, \\ \Omega - (\sqrt{5} - 1) [y - \tfrac{1}{2}(\sqrt{5} + 3)x] &= 0, \\ \Omega - (\sqrt{5} + 3) [y + \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, \\ \Omega - (\sqrt{5} - 1) [y - \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, \\ \Omega + (\sqrt{5} - 1) [y - \tfrac{1}{2}(\sqrt{5} + 3)x] &= 0, \\ \Omega + (\sqrt{5} + 3) [y + \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, \\ \Omega + (\sqrt{5} - 1) [y + \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, \\ \Omega - (\sqrt{5} - 1) [y + \tfrac{1}{2}(\sqrt{5} + 3)x] &= 0, \\ \Omega - (\sqrt{5} + 3) [y - \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, \\ \Omega - (\sqrt{5} - 1) [y + \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, \\ \Omega + (\sqrt{5} - 1) [y + \tfrac{1}{2}(\sqrt{5} + 3)x] &= 0, \\ \Omega + (\sqrt{5} + 3) [y - \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0.\end{aligned}$$

It hence appears that 4 and 4 circles have with  $\Omega = 0$  the common chords  $y + \tfrac{1}{2}(\sqrt{5} - 1)x = 0$ ,  $y - \tfrac{1}{2}(\sqrt{5} - 1)x = 0$  respectively: and that 2 and 2 circles have with  $\Omega = 0$  the common chords  $y + \tfrac{1}{2}(\sqrt{5} + 3)x = 0$ ,  $y - \tfrac{1}{2}(\sqrt{5} + 3)x = 0$  respectively.

**122.** The equations of the 12 circles are, in fact,

$$\begin{aligned}\Omega \pm (\sqrt{5} - 1) [y \pm \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, & \Omega \pm (\sqrt{5} + 3) [y \pm \tfrac{1}{2}(\sqrt{5} - 1)x] &= 0, \\ \Omega \pm (\sqrt{5} - 1) [y \pm \tfrac{1}{2}(\sqrt{5} + 3)x] &= 0;\end{aligned}$$

hence the radii are  $= \sqrt{5} - 1$ , 2 and  $\sqrt{5} + 1$  respectively.

The construction of the 12 circles is as follows. Starting with a circle radius 1.

Lay down the diameters  $y \pm \tfrac{1}{2}(\sqrt{5} - 1)x = 0$  ( $AA$  in the figure), and through the extremities of each describe 2 pairs of circles with the radii  $\sqrt{5} - 1$ ,  $\sqrt{5} + 1$  respectively.

Lay down the diameters  $y \pm \tfrac{1}{2}(\sqrt{5} + 3)x = 0$  ( $BB$  in the figure), and through the extremities of each describe a pair of circles with the radius 2.

**123.** For the *A*-form, the equations of the fifteen planes are at once found to be

$$\begin{aligned}
 y &= 0, & ad \cdot bc \\
 -x + (\epsilon - \epsilon^4)z &= 0, & ac \cdot bd \\
 +(\epsilon - \epsilon^4)x + z &= 0, & ab \cdot cd \\
 (\epsilon - \epsilon^4)x - i(\epsilon^2 + \epsilon^3)y &= 0, & ab \cdot bc \\
 -(\epsilon^2 - \epsilon^3)x + i(\epsilon^2 + \epsilon^3)y + 2(\epsilon + \epsilon^4)z &= 0, & ab \cdot ce \\
 -x + i(\epsilon^2 + \epsilon^3 - \epsilon - \epsilon^4)y + 2z &= 0, & ae \cdot bc \\
 (\epsilon - \epsilon^4)x - i(\epsilon^2 + \epsilon^3)y &= 0, & ab \cdot de \\
 -(\epsilon + \epsilon^4)x + i(\epsilon^2 - \epsilon^3)y + 2(\epsilon - \epsilon^4)z &= 0, & ae \cdot bd \\
 +(\epsilon^2 + \epsilon^3 + 2)x + i(\epsilon^2 - \epsilon^3)y + 2(\epsilon - \epsilon^4)z &= 0, & ad \cdot bc \\
 -(\epsilon - \epsilon^4)x + i(\epsilon + \epsilon^4)y &= 0, & ae \cdot cd \\
 -(\epsilon + \epsilon^4)x - i(\epsilon^2 - \epsilon^3)y + 2(\epsilon + \epsilon^4)z &= 0, & ac \cdot de \\
 +(\epsilon^2 + \epsilon^3 + 2)x + i(\epsilon^2 - \epsilon^3)y + 2z &= 0, & ad \cdot ce \\
 -(\epsilon^2 - \epsilon^3)x + i(\epsilon^2 + \epsilon^3)y &= 0, & bd \cdot ce \\
 +x(\epsilon^2 + \epsilon^3) - i(\epsilon^2 - \epsilon^3)y + 2(\epsilon + \epsilon^4)z &= 0, & bc \cdot de \\
 -x - i(\epsilon^2 + \epsilon^3 - \epsilon - \epsilon^4)y + 2z &= 0, & bc \cdot cd,
 \end{aligned}$$

where, as before, the three planes of each set intersect at right angles.

**124.** Passing to the circles, the first plane of each set gives a right line, and we have thus five of the circles reducing themselves to right lines inclined to the axis of  $x$  at angles  $0^\circ$ ,  $36^\circ$ ,  $72^\circ$ ,  $108^\circ$  and  $144^\circ$  respectively.

The remaining 10 circles form 5 pairs, the circles of a pair having different radii, but the two radii being the same for each pair, and so that for the several pairs the common chords with the circle  $\Omega = 0$  are the diameters inclined to the axis of  $x$  at the angles  $18^\circ$ ,  $54^\circ$ ,  $90^\circ$ ,  $126^\circ$  and  $162^\circ$  respectively. Considering the two circles for which the inclination is  $90^\circ$ , these arise from the planes  $-x + (\epsilon - \epsilon^4)z = 0$ ,  $(\epsilon + \epsilon^4)x + 2z = 0$ , respectively. The equations of the circles thus are  $(\epsilon + \epsilon^4)\Omega + 2x = 0$ ,

$\Omega - 2(\epsilon + \epsilon^4)x = 0$ , or recollecting that  $2(\epsilon + \epsilon^4) = \sqrt{5} - 1$  and therefore  $\frac{2}{\epsilon + \epsilon^4} = \sqrt{5} + 1$ , the equations are

$$x^2 + y^2 - (\sqrt{5} - 1)x - 1 = 0, \quad x^2 + y^2 + (\sqrt{5} + 1)x = 0;$$

hence for the first circle the  $x$ -coordinate of the centre is  $\frac{1}{2}(\sqrt{5} - 1)$  and the radius is  $= \frac{1}{2}\sqrt{10 - 2\sqrt{5}}$ ; for the second circle the  $x$ -coordinate of the centre is  $= \frac{1}{2}(\sqrt{5} + 1)$ , and the radius  $= \frac{1}{2}\sqrt{10 + 2\sqrt{5}}$ . We have thus the construction of these two circles, and consequently the construction of all the 12 circles.

**125.** For the *B*-form, after some easy reductions and attending to the relation  $\omega - \omega^2 = i\sqrt{3}$ , the

equations of the 15 planes become

$$\begin{aligned}
 x &= 0, & ac \cdot bd \\
 (-3 + \sqrt{5})y + 2z &= 0, & ab \cdot cd \\
 (3 + \sqrt{5})y + 2z &= 0, & ad \cdot bc \\
 \sqrt{3}x + \sqrt{5}y + 2z &= 0, & ac \cdot be \\
 -(1 + \sqrt{5})\sqrt{3}x + (3 - \sqrt{5})y + 4z &= 0, & ab \cdot ce \\
 (-1 + \sqrt{5})\sqrt{3}x + (-3 - \sqrt{5})y + 4z &= 0, & ae \cdot bc \\
 x + \sqrt{3}y &= 0, & ae \cdot bd \\
 -\sqrt{3}x + y + (3 + \sqrt{5})z &= 0, & ad \cdot bc \\
 \sqrt{3}x - y + (3 - \sqrt{5})z &= 0, & ab \cdot de \\
 -\sqrt{3}x + \sqrt{5}y + 2z &= 0, & ac \cdot de \\
 (1 - \sqrt{5})\sqrt{3}x + (-3 - \sqrt{5})y + 4z &= 0, & ad \cdot ce \\
 (1 + \sqrt{5})\sqrt{3}x + (3 - \sqrt{5})y + 4z &= 0, & ae \cdot cd \\
 x - \sqrt{3}y &= 0, & bd \cdot ce \\
 \sqrt{3}x + y + (3 + \sqrt{5})z &= 0, & bc \cdot de \\
 -\sqrt{3}x + y + (3 - \sqrt{5})z &= 0, & bc \cdot cd.
 \end{aligned}$$

**126.** Of the 15 circles, 3 are the lines  $x - y\sqrt{3} = 0$ ,  $x = 0$ ,  $x + y\sqrt{3} = 0$ , viz. these are lines at inclinations  $30^\circ$ ,  $90^\circ$ ,  $150^\circ$  to the axis of  $x$ . The equations of the remaining 12 circles are

$$\begin{aligned}
 \Omega + (3 - \sqrt{5})y &= 0, \\
 \Omega - (3 + \sqrt{5})y &= 0, \\
 (3 + \sqrt{5})\Omega - 2(y - x\sqrt{3}) &= 0, \\
 \Omega(3 - \sqrt{5}) + 2(y - x\sqrt{3}) &= 0, \\
 (3 + \sqrt{5})\Omega - 2(y + x\sqrt{3}) &= 0, \\
 (3 - \sqrt{5})\Omega + 2(y + x\sqrt{3}) &= 0,
 \end{aligned}$$

viz. these are pairs of circles having, for their common chords with  $\Omega = 0$ , the diameters at inclinations  $0^\circ$ ,  $60^\circ$ ,  $120^\circ$  respectively. And, lastly, we have the circles

$$\begin{aligned}
 2\Omega - [(-1 + \sqrt{5})\sqrt{3}x - (3 + \sqrt{5})y] &= 0, & 2\Omega + [(-1 + \sqrt{5})\sqrt{3}x + (3 + \sqrt{5})y] &= 0, \\
 \Omega - [-\sqrt{3}x + \sqrt{5}y] &= 0, & \Omega - [\sqrt{3}x + \sqrt{5}y] &= 0, \\
 2\Omega + [(1 + \sqrt{5})\sqrt{3}x - (3 - \sqrt{5})y] &= 0, & 2\Omega - [(1 + \sqrt{5})\sqrt{3}x + (3 - \sqrt{5})y] &= 0.
 \end{aligned}$$

**127.** The first three of these have, for common chords with  $\Omega = 0$ , the diameters whose equations are

$$(-1 + \sqrt{5})\sqrt{3}x - (3 + \sqrt{5})y = 0, \quad -\sqrt{3}x + \sqrt{5}y = 0, \quad (1 + \sqrt{5})\sqrt{3}x - (3 - \sqrt{5})y = 0 :$$

viz. these equations are  $y = (-2 + \sqrt{5})x\sqrt{3}$ ,  $y = \frac{\sqrt{3}}{\sqrt{5}}x$ ,  $y = (2 + \sqrt{5})x\sqrt{3}$ . If, as in a foregoing table,  $\theta = 37^\circ 46'$ ,  $\sin \theta = \sqrt{\frac{5}{45}}$ ,  $\cos \theta = \sqrt{\frac{3}{45}}$ , and therefore  $\tan \theta = \sqrt{\frac{5}{3}}$ ; then the inclinations of these diameters to the axis of  $x$  are respectively  $60^\circ - \theta$ ,  $\theta$  and  $120^\circ - \theta$ , or say  $30^\circ - (\theta - 30^\circ)$ ,  $30^\circ + (\theta - 30^\circ)$  and  $90^\circ - (\theta - 30^\circ)$ , where  $\theta - 30^\circ = 7^\circ 46'$ , i.e. the inclinations are  $30^\circ \pm 7^\circ 46'$  and  $90^\circ - 7^\circ 46'$ . And for the other three circles the common chords are the diameters at the same inclinations taken negatively. The geometrical construction of the fifteen circles for the  $B$ -case in question is thus not so simple as in the  $\Theta$ - and  $A$ -cases.

*The Regular Polyhedra as Solid figures. Art. Nos. 128 to 134.*

**128.** I annex some results relating to the polyhedra considered as solid figures bounded by plane faces; or say results relating to the regular solids:  $s$  is in each case taken for the length of the edge of the solid.

|                                        | Tetrahedron                                   | Cube                         | Octahedron                                        | Dodecahedron                             | Icosahedron                            |
|----------------------------------------|-----------------------------------------------|------------------------------|---------------------------------------------------|------------------------------------------|----------------------------------------|
| Edge                                   | $s$                                           | $s$                          | $s$                                               | $s$                                      | $s$                                    |
| Rad. of circum. sphere, $R$            | $s \cdot \frac{\sqrt{3}}{2\sqrt{2}}$          | $s \cdot \frac{\sqrt{3}}{2}$ | $s \cdot \frac{1}{\sqrt{2}}$                      | $s \cdot \frac{\sqrt{3}(\sqrt{5}+1)}{4}$ | $s \sqrt{\frac{5+\sqrt{5}}{8}}$        |
| Rad. of inters. sphere, $\rho$         | $s \cdot \frac{1}{2\sqrt{2}}$                 | $s \cdot \frac{1}{\sqrt{2}}$ | $s \cdot \frac{1}{2}$                             | $s \cdot \frac{3+\sqrt{5}}{4}$           | $s \cdot \frac{1+\sqrt{5}}{4}$         |
| Rad. of inscribed sphere, $r$          | $s \cdot \frac{1}{2\sqrt{2}\sqrt{3}}$         | $s \cdot \frac{1}{2}$        | $s \cdot \frac{1}{\sqrt{6}}$                      | $s \cdot \sqrt{\frac{5+2\sqrt{5}}{20}}$  | $s \cdot \frac{3+\sqrt{5}}{4\sqrt{3}}$ |
| Rad. of circle circum. to face, $R'$   | $s \cdot \frac{1}{\sqrt{3}}$                  | $s \cdot \frac{1}{\sqrt{2}}$ | $s \cdot \frac{1}{\sqrt{3}}$                      | $s \cdot \sqrt{\frac{5+\sqrt{5}}{10}}$   | $s \cdot \frac{1}{\sqrt{3}}$           |
| Rad. of circle inscribed to face, $r'$ | $s \cdot \frac{1}{2\sqrt{3}}$                 | $s \cdot \frac{1}{2}$        | $s \cdot \frac{1}{2\sqrt{3}}$                     | $s \cdot \sqrt{\frac{5+2\sqrt{5}}{20}}$  | $s \cdot \frac{1}{2\sqrt{3}}$          |
| Incl. of adjacent faces                | $\cos^{-1} \frac{1}{3} = 70^\circ 28'$        | $90^\circ$                   | $\cos^{-1} -\frac{1}{3} = 109^\circ 32'$          |                                          |                                        |
| Incl. of edge to adjacent face         | $\cos^{-1} \frac{1}{\sqrt{3}} = 54^\circ 46'$ | $90^\circ$                   | $= \cos^{-1} -\frac{1}{\sqrt{3}} = 125^\circ 44'$ |                                          |                                        |

But we require further data in the cases of the dodecahedron and the icosahedron respectively.

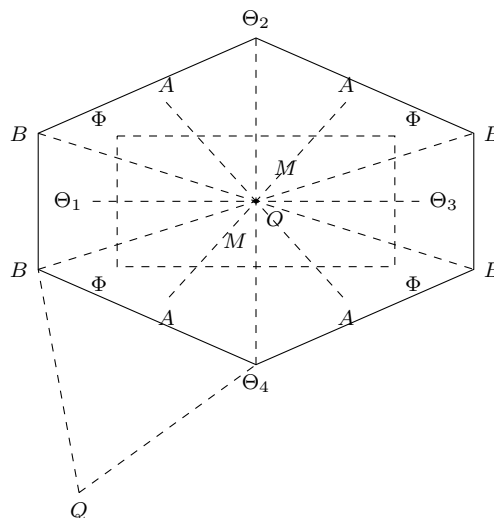
**129.** For the dodecahedron, taking the edge to be  $= s$  as before, then in the pentagonal face

$$\begin{aligned} \text{diagonal, } g \text{ is } &= s \cdot \frac{1}{2}(\sqrt{5} + 1), \\ \text{altitude, } k, &= s \cdot \frac{1}{2}\sqrt{5 + 2\sqrt{5}}, \\ \text{segments of do., } e, &= s \cdot \frac{1}{4}\sqrt{10 - 2\sqrt{5}}, \\ f, &= s \cdot \frac{1}{4}\sqrt{10 + 2\sqrt{5}}, \end{aligned}$$

where

$$k = e + f = R' + r'.$$

**130.** The section through a pair of opposite edges is a hexagon, as shown in the figure, viz. this is constructed by taking the four equal distances  $O\Theta_i = \rho$ ,  $= s \cdot \frac{1}{4}(3 + \sqrt{5})$ , meeting at right angles in  $O$ ; then drawing the double ordinates  $BB$ , each  $= s$ , through  $\Theta_1$  and  $\Theta_3$  respectively, and joining their extremities with  $\Theta_2$  and  $\Theta_4$ : the sides  $\Theta_2B$  and  $\Theta_4B$  are then each  $= k$ ,  $= s \cdot \frac{1}{2}\sqrt{5 + 2\sqrt{5}}$ ; and inserting upon them the points  $A$ ,  $\Phi$  from the figure of the pentagon, we have several geometrical relations; viz. the line  $AA$  cuts the parallel sides  $B\Theta_2$ ,  $B\Theta_4$  at right angles, and when produced passes through the intersection of  $B\Theta_1$  and  $B\Theta_3$ : we have  $OA$ ,  $OB$ ,  $O\Theta = r$ ,  $R$ ,  $\rho$  respectively: the four points  $\Phi$  form a square, the side of which is  $g$ ,  $= s \cdot \frac{1}{2}(\sqrt{5} + 1)$ .

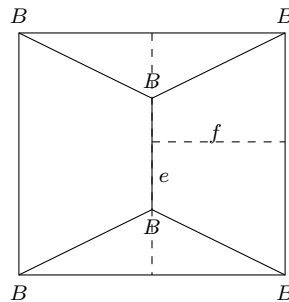


**131.** We find also

$$\begin{aligned} AM &= s\sqrt{\frac{5+\sqrt{5}}{10}}, \\ AQ &= s\sqrt{\frac{3+2\sqrt{3}}{5}}, \\ QM &= s\sqrt{\frac{25+11\sqrt{5}}{5}} = r \cdot 2\sqrt{2}, \\ OQ &= s\sqrt{\frac{25+11\sqrt{5}}{8}} = r\sqrt{5}, \\ OM &= s\sqrt{\frac{5-\sqrt{5}}{40}}, \\ MB &= s\sqrt{\frac{2(5+2\sqrt{5})}{8}}. \end{aligned}$$

It may be remarked that in the figure  $B\Theta_2$ ,  $B\Theta_4$  are the projections of pentagonal faces, at right angles to the plane of the paper, having their centres at the points  $A$ ,  $A$ , and the perpendicular distance between them =  $AA$ : the points  $Q$ ,  $Q$  (only one of them shown in the figure) determine the directions of the  $5 + 5$  sides which abut on these pentagonal faces respectively; and the  $5 + 5$  points  $B$  which are the other extremities of these sides respectively form two pentagons, centres  $M$ ,  $M$ , in the planes  $MB MB$  respectively: the remaining 10 sides of the dodecahedron are the skew decagon obtained by joining in order these 10 points  $B$ . We have thus the means of making the perspective delineation of the dodecahedron.

**132.** The dodecahedron is built up from the cube, by placing on each face a figure of two triangular and two quadrangular faces, the orthogonal projection of which on the face of the cube is as in the figure: the side of the square is  $g$ ,  $= s \cdot \frac{1}{2}(\sqrt{5} + 1)$ : the slope-breadths of the triangular faces are  $e$ ,  $= s \cdot \frac{1}{4}\sqrt{10 - 2\sqrt{5}}$ , and those of the quadrangular faces are  $f$ ,  $= s \cdot \frac{1}{4}\sqrt{10 + 2\sqrt{5}}$ ; the lines represented by the other lines of the figure are in actual length each  $= s$ . We have thus a



section which is an isosceles triangle, base  $= g$ , other sides each  $= f$ ; and the square of the altitude is thus  $= f^2 - \frac{1}{4}g^2 = \frac{1}{4}s^2$ , or the altitude  $= \frac{1}{2}s$ ; viz. the altitude of the ridge-line  $BB$ , above the face of the cube is  $= \frac{1}{2}s$ , the half-side of the dodecahedron. We have in this result the most simple means of forming the perspective delineation of the dodecahedron.

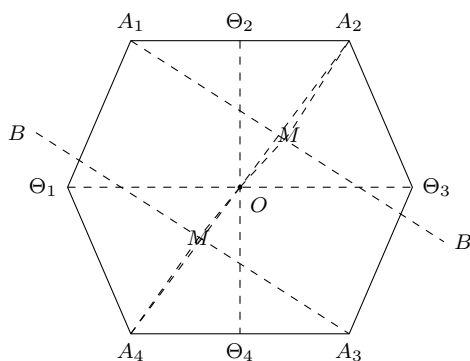
**133.** For the icosahedron the section through two opposite edges is a hexagon, as shown in the figure (p. 216): to construct it, we take the four distances  $O\Theta$  each  $= \rho$ ,  $= s \cdot \frac{1}{4}(1 + \sqrt{5})$  meeting at right angles; and then the distances  $A\Theta_2$ ,  $A\Theta_4$  each  $= \frac{1}{2}s$ ; and complete the hexagon. This gives the sides  $A\Theta_1$ ,  $A\Theta_3$  each  $= s \cdot \frac{1}{2}\sqrt{3}$ , the altitude of the triangular face, side  $= s$ ; and then, taking  $\Theta_2B$  one-third of this,  $= \frac{s\sqrt{3}}{6}$ , we have  $OB$  at right angles to  $A\Theta_2$ , and  $OA$ ,  $OB$ ,  $O\Theta = R$ ,  $r$ ,  $\rho$  respectively.

Moreover, joining  $A\Theta_3$  and  $OA_1$ , we have these lines cutting at right angles in a point  $M$ : we find

$$\begin{aligned} A_1\Theta_3 &= s \cdot \frac{1}{2}\sqrt{5 + 2\sqrt{5}}, \\ M\Theta_3 &= s\sqrt{\frac{5+2\sqrt{5}}{20}}, \\ A_1M &= s\sqrt{\frac{5+\sqrt{5}}{10}}, \\ OM &= s\sqrt{\frac{3+\sqrt{5}}{40}} = \frac{1}{2}A_1M, \\ A_1M &= s\sqrt{\frac{5-\sqrt{5}}{10}}. \end{aligned}$$

**134.** It may be remarked that  $A_1\Theta_3$ ,  $A_3\Theta_1$  are the projections of two pentagons in planes perpendicular to that of the paper, their centres being  $M$ ,  $M'$ : producing  $OM$ ,  $OM'$  to the points  $A_1$ ,  $A_3$  respectively, we have a pentagonal pyramid, summit  $A_1$ , standing on the first pentagon, and an opposite pyramid, summit  $A_3$ , standing on

the other pentagon: the  $5 + 5$  triangular faces of the two pyramids are ten of the faces of the icosahedron, and the remaining ten faces are the triangles each having for its base a side of the one pentagon, and for its vertex a summit of the other pentagon, viz. the sides are the sides of the skew decagon obtained by joining in order the angular points of the two pentagons. We have thus a convenient method of forming the perspective delineation of the icosahedron.



746.

#### HIGHER PLANE CURVES.

[From Salmon's *Higher Plane Curves*, (3rd ed., 1879); see the Preface.]

One chapter and a large number of articles, in the second edition of Salmon's *Higher Plane Curves*, are due to Professor Cayley. Full reference to these is given by Dr Salmon in the preface.

747.

#### NOTE ON THE DEGENERATE FORMS OF CURVES.

[From Salmon's *Higher Plane Curves*, (3rd ed., 1879), pp. 383–385.]

Some remarks may be added as to the analytical theory of the degenerate forms of curves. As regards conics, a line-pair can be represented in point-coordinates by an equation of the form  $xy = 0$ ; and reciprocally a point-pair can be represented in line-coordinates by an equation  $\xi\eta = 0$ , but we have to consider how the point-pair can be represented in point-coordinates: an equation  $x^2 = 0$  is no adequate representation of the point-pair, but merely represents (as a two-fold or twice repeated line) the line joining the two points of the point-pair, all traces of the points themselves being lost in this representation: and it is to be noticed, that the conic, or two-fold line  $x^2 = 0$ , or say  $(\alpha x + \beta y + \gamma z)^2 = 0$  is a conic which, analytically, and (in an improper sense) geometrically, satisfies the condition of touching any line whatever; whereas the only proper tangents of a point-pair are the lines which pass through one or other of the two points of the point-pair.

The solution arises out of the notion of a point-pair, considered as the limit of a conic, or say as an indefinitely flat conic; we have to consider conics certain of the coefficients whereof are infinitesimals, and which, when the infinitesimal coefficients actually vanish, reduce themselves to two-fold lines; and it is, moreover, necessary to consider the evanescent coefficients as infinitesimals of different orders. Thus consider the conics which pass through two given points, and touch two given lines (four conditions); take

$y = 0$ ,  $z = 0$  for the given lines,  $x = 0$  for the line joining the given points, and  $(x = 0, y - \alpha z = 0)$ ,  $(x = 0, y - \beta z = 0)$  for the given points; the equation of a conic satisfying the required conditions and containing one arbitrary parameter  $\theta$ , is

$$x^2 + 2\theta xy + 2\theta\sqrt{(\alpha\beta)}xz + \theta^2(y - \alpha z)(y - \beta z) = 0;$$

or, what is the same thing,

$$\{x + \theta y + \theta\sqrt{(\alpha\beta)}z\}^2 - \theta^2(\alpha + \beta)yz = 0;$$

and this equation, considering therein  $\theta$  as an infinitesimal, say of the first order, represents the flat conic or point-pair composed of the two given points. Comparing with the general equation

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

we have

$$a = 1, \quad b = \theta^2, \quad c = \theta^2\alpha\beta, \quad f = -\frac{1}{2}\theta^2(\alpha + \beta), \quad g = \frac{1}{2}\theta\sqrt{(\alpha\beta)}, \quad h = \theta,$$

viz.  $a$  being taken to be finite, we have  $g$  and  $h$  infinitesimals of the first order,  $b, c, f$  infinitesimals of the second order; and the four ratios  $g : \sqrt{(b)} : \sqrt{(c)} : \sqrt{(f)} : h$  are so determined as to satisfy the prescribed conditions.

Observe that the flat conic, considered as a conic passing through the two given points and touching the two given lines, is represented by a determinate equation, viz. considering the condition imposed upon ( $b = \text{infinitesimal}$ ) as a determination of  $\theta$ , the equation is a completely determinate one; but considering the flat conic merely as a conic passing through the two given points, the equation would contain two arbitrary parameters, determinable if the flat conic was subjected to the condition of touching two given lines, or to any other two conditions.

Generally, we may consider the equation of a curve of the order  $n$ ; such equation containing certain infinitesimal coefficients and, when these vanish, reducing itself to a composite equation  $= P^\alpha Q^\beta \dots$ ; the equation in its original form represents a curve which may be called the penultimate curve. Consider the tangents from an arbitrary point to the penultimate curve; when this breaks up, the system of tangents reduces itself to (1) the tangents from the fixed point to the several component curves  $P = 0$ ,  $Q = 0$ , &c. respectively; (2) the lines through the singular points of these same curves respectively; (3) the lines through the points of intersection of each two of the component curves; these points, each reckoned a proper number of times, are called “fixed summits”; (4) the lines from the fixed point to certain determinate points called “free summits” on the several component curves  $P = 0$ ,  $Q = 0$ , &c. respectively. We have thus a degenerate form of the  $n$ -thic curve, which may be regarded as consisting of the component curves, each its proper number of times, and of the foregoing points called summits, and is consequently only inadequately represented by the ultimate equation  $= P^\alpha Q^\beta \dots$ ; the number and distribution of the summits is not arbitrary, but is regulated by laws arising from the consideration of the penultimate curve, and there are of course for any given value of  $n$  various forms of degenerate curve, according to the different ultimate forms  $= P^\alpha Q^\beta \dots = 0$ , and to the number and distribution of the summits on the different component curves. The case of a quartic curve having the ultimate form  $x^2y^2$  has been considered by Cayley, *Comptes Rendus*, t. LXXIV. p. 708 (March, 1872), [515], who states his conclusion as follows:

“there exists a quartic curve the penultimate of  $= x^2y^2 = 0$ , with nine free summits, three of them on one of the lines (say the line  $y = 0$ ), and which are three of the intersections of the quartic by this line (the fourth intersection being indefinitely near to the point  $x = 0, y = 0$ ), six situate at pleasure on the other line  $x = 0$ ; and three fixed summits at the intersection of the two lines.” Other forms have been considered by Dr Zeuthen, *Comptes Rendus*, t. LXXV. pp. 703 and 950 (September and October, 1872), and some other forms by Zeuthen; the whole question of the degenerate forms of curves is one well deserving further investigation.

The question of the number of cubic curves satisfying given elementary conditions (depending as it does on the consideration of the degenerate forms of these curves) has been solved by Maillard and Zeuthen; that of the number of quartic curves has been solved by Dr Zeuthen.

748.

# ON THE BITANGENTS OF A QUARTIC.

[From Salmon's *Higher Plane Curves*, (3rd ed., 1879), pp. 387–389.]

The equations of the 28 bitangents of a quartic curve were obtained in a very elegant form by Riemann in the paper “Zur Theorie der Abel’schen Functionen für den Fall  $p = 3$ ,” *Ges. Werke*, Leipzig, 1876, pp. 456–472; and see also Weber’s *Theorie der Abel’schen Functionen vom Geschlecht 3*, Berlin, 1876. Riemann connects the several bitangents with the characteristics of the 28 odd functions, thus obtaining for them an algorithm which it is worth while to explain, but they will be given also with the algorithm employed p. 231 *et seq.* of the present work\*, which is in fact the more simple one. The characteristic of a triple  $\vartheta$ -function is a symbol of the form

$$\begin{matrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \end{matrix}$$

where each of the letters is = 0 or 1; there are thus in all 64 such symbols, but they are considered as odd or even according as the sum  $\alpha\alpha' + \beta\beta' + \gamma\gamma'$  is odd or even; and the numbers of the odd and even characteristics are 28 and 36 respectively; and, as already mentioned, the 28 odd characteristics correspond to the 28 bitangents respectively.

We have  $x, y, z$  trilinear coordinates,  $\alpha, \beta, \gamma, \alpha', \beta', \gamma'$  constants chosen at pleasure, and then  $\alpha'', \beta'', \gamma''$  determinate constants, such that the equations

$$\begin{aligned} \alpha x + \beta y + \gamma z + \frac{\xi}{\alpha} + \frac{\eta}{\beta} + \frac{\zeta}{\gamma} &= 0, \\ \alpha' x + \beta' y + \gamma' z + \frac{\xi}{\alpha'} + \frac{\eta}{\beta'} + \frac{\zeta}{\gamma'} &= 0, \\ \alpha'' x + \beta'' y + \gamma'' z + \frac{\xi}{\alpha''} + \frac{\eta}{\beta''} + \frac{\zeta}{\gamma''} &= 0, \end{aligned}$$

[\* That is, Salmon’s *Higher Plane Curves*.]

are equivalent to three independent equations; this being so, they determine  $\xi, \eta, \zeta$ , each of them as a linear function of  $(x, y, z)$ ; and the equations of the bitangents of the curve  $\sqrt{(x\xi)} + \sqrt{(y\eta)} + \sqrt{(z\zeta)} = 0$  (see Weber, p. 100) are

|    |                   |                                                                                                                                             |
|----|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 18 | $\frac{111}{111}$ | $x = 0,$                                                                                                                                    |
| 28 | $\frac{001}{011}$ | $y = 0,$                                                                                                                                    |
| 38 | $\frac{011}{001}$ | $z = 0,$                                                                                                                                    |
| 23 | $\frac{010}{010}$ | $\xi = 0,$                                                                                                                                  |
| 13 | $\frac{100}{110}$ | $\eta = 0,$                                                                                                                                 |
| 12 | $\frac{110}{100}$ | $\zeta = 0,$                                                                                                                                |
| 48 | $\frac{101}{100}$ | $x + y + z = 0,$                                                                                                                            |
| 14 | $\frac{010}{011}$ | $\xi + y + z = 0,$                                                                                                                          |
| 58 | $\frac{100}{101}$ | $\alpha x + \beta y + \gamma z = 0,$                                                                                                        |
| 15 | $\frac{011}{010}$ | $\frac{\xi}{\alpha} + \beta y + \gamma z = 0,$                                                                                              |
| 68 | $\frac{110}{010}$ |                                                                                                                                             |
| 16 | $\frac{001}{101}$ |                                                                                                                                             |
| 78 | $\frac{010}{110}$ | $\alpha'' x + \beta'' y + \gamma'' z = 0,$                                                                                                  |
| 17 | $\frac{101}{001}$ | $\frac{\xi}{\alpha} + \beta'' y + \gamma'' z = 0,$                                                                                          |
| 24 | $\frac{100}{111}$ | $x + \eta + z = 0,$                                                                                                                         |
| 34 | $\frac{110}{101}$ | $x + y + \zeta = 0,$                                                                                                                        |
| 25 | $\frac{101}{110}$ | $\alpha x + \frac{\eta}{\beta} + \gamma z = 0,$                                                                                             |
| 35 | $\frac{111}{100}$ | $\alpha x + \beta y + \frac{\zeta}{\gamma} = 0,$                                                                                            |
| 26 | $\frac{111}{001}$ | $\alpha' x + \frac{\eta}{\beta'} + \gamma' z = 0,$                                                                                          |
| 36 | $\frac{101}{011}$ | $\alpha' x + \beta' y + \frac{\zeta}{\gamma'} = 0,$                                                                                         |
| 27 | $\frac{011}{101}$ | $\alpha'' x + \frac{\eta}{\beta''} + \gamma'' z = 0,$                                                                                       |
| 37 | $\frac{001}{111}$ | $\alpha'' x + \beta'' y + \frac{\zeta}{\gamma''} = 0,$                                                                                      |
| 67 | $\frac{100}{100}$ | $\frac{x}{1 - \beta\gamma} + \frac{y}{1 - \gamma\alpha} + \frac{z}{1 - \alpha\beta} = 0,$                                                   |
| 57 | $\frac{110}{011}$ | $\frac{x}{1 - \beta'\gamma'} + \frac{y}{1 - \gamma'\alpha'} + \frac{z}{1 - \alpha'\beta'} = 0,$                                             |
| 56 | $\frac{010}{111}$ | $\frac{x}{1 - \beta''\gamma''} + \frac{y}{1 - \gamma''\alpha''} + \frac{z}{1 - \alpha''\beta''} = 0,$                                       |
| 45 | $\frac{001}{001}$ | $\frac{\xi}{\alpha(1 - \beta\gamma)} + \frac{\eta}{\beta(1 - \gamma\alpha)} + \frac{\zeta}{\gamma(1 - \alpha\beta)} = 0,$                   |
| 46 | $\frac{011}{110}$ | $\frac{\xi}{\alpha'(1 - \beta'\gamma')} + \frac{\eta}{\beta'(1 - \gamma'\alpha')} + \frac{\zeta}{\gamma'(1 - \alpha'\beta')} = 0,$          |
| 47 | $\frac{111}{010}$ | $\frac{\xi}{\alpha''(1 - \beta''\gamma'')} + \frac{\eta}{\beta''(1 - \gamma''\alpha'')} + \frac{\zeta}{\gamma''(1 - \alpha''\beta'')} = 0.$ |

The whole number of ways in which the equation of the curve can be expressed in a form such as  $\sqrt{(x\xi)} + \sqrt{(y\eta)} + \sqrt{(z\zeta)} = 0$  is 1260; viz. the three pairs of bitangents entering into the equation of the curve are of one of the types

|        |        |       |        |       |
|--------|--------|-------|--------|-------|
| 12.34, | 13.24, | 14.23 | No. is | 70    |
| 12.34, | 13.24, | 56.78 | „      | 630   |
| 13.23, | 14.24, | 15.25 | „      | 560   |
|        |        |       |        | 1260. |

It may be remarked that, selecting at pleasure any two pairs out of a system of three pairs, the type is always  $\square$  or  $|||$ , viz. (see p. 233) the four bitangents are such that their points of contact are situate on a conic.

A considerable number of articles in the third edition of Salmon's *Treatise* are due to Professor Cayley. Full reference to these is given by Dr Salmon in the preface.

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750.

# ON THE THEORY OF RECIPROCAL SURFACES.

[From Salmon's *Treatise* on the analytic geometry of three dimensions, (3rd ed., 1874), pp. 539–550.]

**600.** In further developing the theory of reciprocal surfaces it has been found necessary to take account of other singularities, some of which are as yet only imperfectly understood. It will be convenient to give the following complete list of the quantities which present themselves:

- $n$ , order of the surface.
- $a$ , order of the tangent cone drawn from any point to the surface.
- $h$ , number of nodal edges of the cone.
- $\kappa$ , number of its cuspidal edges.
- $\rho$ , class of nodal torse.
- $\sigma$ , class of cuspidal torse.
- $b$ , order of nodal curve.
- $k$ , number of its apparent double points.
- $f$ , number of its actual double points.
- $t$ , number of its triple points.
- $j$ , number of its pinch-points.
- $q$ , its class.
- $c$ , order of cuspidal curve.
- $h$ , number of its apparent double points.
- $\theta$ , number of its points of an unexplained singularity.
- $\chi$ , number of its close-points.
  
- $\omega$ , number of its off-points.
- $r$ , its class.
- $\beta$ , number of intersections of nodal and cuspidal curves, stationary points on cuspidal curve.
- $\gamma$ , number of intersections, stationary points on nodal curve.
- $i$ , number of intersections, not stationary points on either curve.
- $C$ , number of cnicnodes of surface.
- $B$ , number of binodes.

And corresponding reciprocally to these:

- $n'$ , class of surface.
- $a'$ , class of section by arbitrary plane.
- $\delta'$ , number of double tangents of section.

$\kappa'$ , number of its inflexions.

$\rho'$ , order of node-couple curve.

$\sigma'$ , order of spinode curve.

$b'$ , class of node-couple torse.

$k'$ , number of its apparent double planes.

$f'$ , number of its actual double planes.

$t'$ , number of its triple planes.

$j'$ , number of its pinch-planes.

$q'$ , its order.

$c'$ , class of spinode torse.

$h'$ , number of its apparent double planes.

$\theta'$ , number of its planes of a certain unexplained singularity.

$\chi'$ , number of its close-planes.

$\omega'$ , number of its off-planes.

$r'$ , its order.

$\beta'$ , number of common planes of node-couple and spinode torse, stationary planes of spinode torse.

$\gamma'$ , number of common planes, stationary planes of node-couple torse.

$i'$ , number of common planes, not stationary planes of either torse.

$C'$ , number of cnictropes of surface.

$B'$ , number of its bitropes.

In all, these are 46 quantities.

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[PAPER 750 CONTINUES BEYOND P. 226 WITH ART. 601 GIVING THE RELATIONS AMONG THE LISTED QUANTITIES; SEE THE NEXT SLICE.]